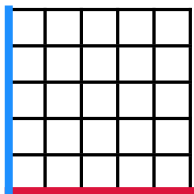


Geometry induced by representation mappings

ReLU activation

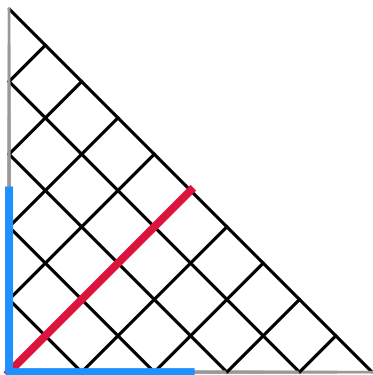
Identity map
(pre-activation W)



$$W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda_1 = 1, \lambda_2 = 1$$

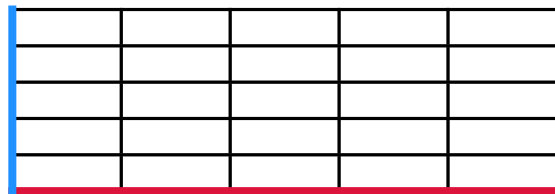
Complex eigenvalues
(pre-activation W)



$$W = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\lambda_1 = 1 - 1i, \lambda_2 = 1 + 1i$$

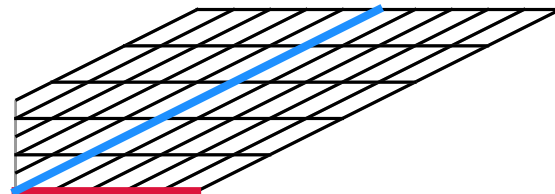
Distinct real eigenvalues
(anisotropic scaling)



$$W = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda_1 = 1, \lambda_2 = 3$$

Repeated eigenvalue
(shear, one eigendirection)



$$W = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\lambda_1 = 1, \lambda_2 = 1$$